System Requirements Specification

for

Microwave Tracking Ground Station

Version 1.0 approved

Prepared by <Jusefs Saade; Christian Ogburn; Jared O'Shea; Ian Wolfe; Clay Edwards; Jordan West>

<WiDE LAB>

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Revision History

Name	Date	Reason For Changes	Version
Christian Ogburn	09/28/2023	Document Creation	1

1. Introduction

1.1 Purpose

<Identify the product whose software requirements are specified in this document, including the revision or release number. Describe the scope of the product that is covered by this SRS, particularly if this SRS describes only part of the system or a single subsystem.>

1.2 Document Conventions

<Describe any standards or typographical conventions that were followed when writing this SRS, such as fonts or highlighting that have special significance. For example, state whether priorities for higher-level requirements are assumed to be inherited by detailed requirements, or whether every requirement statement is to have its own priority.>

1.3 Intended Audience and Reading Suggestions

The Microwave Tracking Ground Station is intended for Dr. Eduardo Rojas, Director of the WiDE (Wireless Devices and Electronics) laboratory, and for future students involved in microwave tracking design projects.
<Describe the different types of reader that the document is intended for, such as developers, project managers, marketing staff, users, testers, and documentation writers. Describe what the rest of this SRS contains and how it is organized. Suggest a sequence for reading the document, beginning with the overview sections and proceeding through the sections that are most pertinent to each reader type.>

1.4 Product Scope

The final iteration of this product is expected to be able to track a satellite in orbit and receive data from said satellite. The product will have an antenna fixed to a positioning system, controlled via software to track a satellite as it orbits the Earth. There will also be a graphical interface element.

<Provide a short description of the software being specified and its purpose, including relevant benefits, objectives, and goals. Relate the software to corporate goals or business strategies. If a separate vision and scope document is available, refer to it rather than duplicating its contents here.>

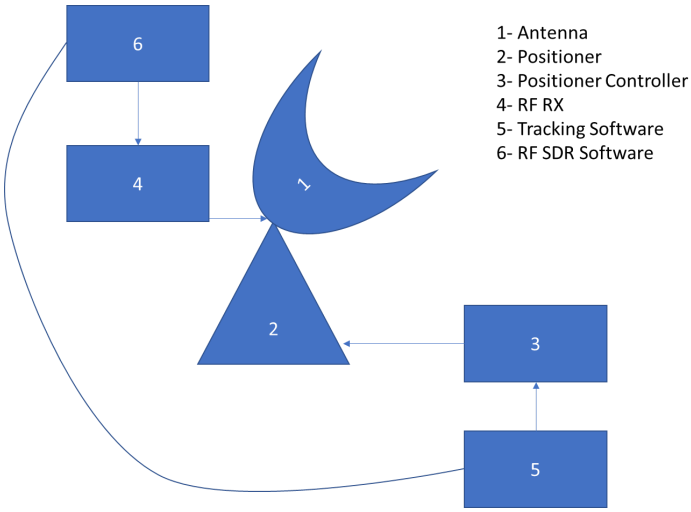
1.5 References

None currently.
<List any other documents or Web addresses to which this SRS refers. These may include user interface style guides, contracts, standards, system requirements specifications, use case documents, or a vision and scope document. Provide enough information so that the reader could access a copy of each reference, including title, author, version number, date, and source or location.>

2. Overall Description

2.1 Product Perspective

This product is intended to build up and eventually replace the product made by the originators of this project last year. The original design from the previous year was able to do basic movement function, but had gain issues and was unable to get signals from satellites.



<Describe the context and origin of the product being specified in this SRS. For example, state whether this product is a follow-on member of a product family, a replacement for certain existing systems, or a new, self-contained product. If the SRS defines a component of a larger system, relate the requirements of the larger system to the functionality of this software and identify interfaces between the two. A simple diagram that shows the major components of the overall system, subsystem interconnections, and external interfaces can be helpful.>

2.2 Product Functions

- The ground station must be able to receive signals from a satellite
- The ground station must be able to detect signals in the S-Band frequency range
- The ground station must be portable and easy to move (smaller than existing prototype in WiDE lab)
- The ground station's dish must be able to turn in all directions in a hemispherical range of motion
- The ground station must be able to be controlled by directly connected controls as well as remotely by computer software
- The ground station must be able to track a specific satellite to receive constant signals

2.3 User Classes and Characteristics

This product's main users and interested audiences would be the Dr. Rojas, members of the IEEE community, former and future peers in electrical engineering, members of the WiDE Lab. Generally, we are expecting our users to be knowledgeable and relatively experienced in the areas of RF and Satellite Tracking.

<Identify the various user classes that you anticipate will use this product. User classes may be differentiated based on frequency of use, subset of product functions used, technical expertise, security or privilege levels, educational level, or experience. Describe the pertinent characteristics of each user class. Certain requirements may pertain only to certain user classes. Distinguish the most important user classes for this product from those who are less important to satisfy.>

2.4 Operating Environment

The tracking ground station is intended to operate outdoors to minimize interference from transmissions such as WiFi and Bluetooth, which would have higher concentrations indoors. Preferable operation locations may include the roof of the MicaPlex and the Lehman building. Additional design considerations will need to be given to mitigate any harm to the product from the elements, including but not limited to wind, humidity, uncontrolled climate (heat and cold), and direct sunlight/UV exposure. With this product being developed in Florida, rain is a major concern for outdoor operation, and ideally such exposure should be avoided at all costs rather than developing the product with water resistance in mind, as such exposure would likely result in catastrophic setbacks.
<Describe the environment in which the software will operate, including the hardware platform, operating system and versions, and any other software components or applications with which it must peacefully coexist.>

2.5 Design and Implementation Constraints

Based on our setup and design of antenna, SDR and operating frequency will limit options on what signals to receive and from how far.
The actuators/motors must align the antenna with the moving satellite in LEO.
The
<Describe any items or issues that will limit the options available to the developers. These might include: corporate or regulatory policies; hardware limitations (timing requirements, memory requirements); interfaces to other applications; specific technologies, tools, and databases to be used; parallel operations; language requirements; communications protocols; security considerations; design conventions or programming standards (for example, if the customer's organization will be responsible for maintaining the delivered software).>

2.6 User Documentation

We are currently unable to complete this section, as we have not been given specifics by the product owner of certain hardware/software that is meant to be used in the design.

<List the user documentation components (such as user manuals, on-line help, and tutorials) that will be delivered along with the software. Identify any known user documentation delivery formats or standards.>

2.7 Assumptions and Dependencies

- Miniaturizing part of the prototype may be difficult due to the use of available motors and parts
- Timeline may affect progress due to conflict with meeting times

- Reused software from previous team may be difficult to incorporate or redesign with new prototype
- The use of a new dish or designing a new one to accomplish the goals of this SRS may prove to be difficult with available parts

<List any assumed factors (as opposed to known facts) that could affect the requirements stated in the SRS. These could include third-party or commercial components that you plan to use, issues around the development or operating environment, or constraints. The project could be affected if these assumptions are incorrect, are not shared, or change. Also identify any dependencies the project has on external factors, such as software components that you intend to reuse from another project, unless they are already documented elsewhere (for example, in the vision and scope document or the project plan).>

3. External Interface Requirements

3.1 User Interfaces

Not adding actual requirements yet as not enough details have been established
This device primarily connects to a computer digitally, although there are some physical aspects to consider. To enhance its portability and ease of maneuvering, it is affixed to a cart, significantly reducing the effort required for relocation. The digital interface encompasses both the SDR software and the positioner control software, both accessible via a computer. These software applications will be developed with a strong emphasis on user-friendly design.

<Describe the logical characteristics of each interface between the software product and the users. This may include sample screen images, any GUI standards or product family style guides that are to be followed, screen layout constraints, standard buttons and functions (e.g., help) that will appear on every screen, keyboard shortcuts, error message display standards, and so on. Define the software components for which a user interface is needed. Details of the user interface design should be documented in a separate user interface specification.>

3.2 Hardware Interfaces

Not adding actual requirements yet as not enough details have been established
The hardware interface requirements for this system comprise two key communication pathways: the antenna feed and the positioner control. The antenna feed is sourced directly from the dish antenna, featuring separate vertical and horizontal polarization ports. These signals are amalgamated into a circularly polarized signal, which subsequently passes through a sequence of filters. The resultant signal is then directed into the Software-Defined Radio (SDR) for interpretation and visualization via the SDR software. Conversely, the positioner control signal is transmitted from the control software to an auxiliary module integrated with the motor controller. This specialized module is inherently designed to seamlessly interface with the motor controller, facilitating computer-based control. Both pathways require interfacing with an external computer that can run the software.

<Describe the logical and physical characteristics of each interface between the software product and the hardware components of the system. This may include the supported device types, the

nature of the data and control interactions between the software and the hardware, and communication protocols to be used.>

3.3 Software Interfaces

Not adding actual requirements yet as not enough details have been established
The software interface requirements for this product revolve around two primary software components. The initial software component interacts with the positioner controller, ensuring precise alignment of the antenna with moving satellites. This software relies on input data, specifically the Keplerian elements of the satellite and the receiver's location, to regulate the positioner and achieve antenna alignment. The second software component is the SDR receiver software, which serves as the intermediary between the system's Software-Defined Radio (SDR) and a computer. This software is responsible for visualizing the signals received by the antenna and fine-tuning the frequencies monitored by the SDR.

<Describe the connections between this product and other specific software components (name and version), including databases, operating systems, tools, libraries, and integrated commercial components. Identify the data items or messages coming into the system and going out and describe the purpose of each. Describe the services needed and the nature of communications. Refer to documents that describe detailed application programming interface protocols. Identify data that will be shared across software components. If the data sharing mechanism must be implemented in a specific way (for example, use of a global data area in a multitasking operating system), specify this as an implementation constraint.>

3.4 Communications Interfaces

Not adding actual requirements yet as not enough details have been established
The communication interface requirements for this system encompass three critical components. Firstly, the system must support both vertical and horizontal polarization modes, combine them into a circularly polarized signal, and apply necessary signal processing filters before transmitting to the Software-Defined Radio (SDR) for further analysis. Secondly, a robust interface is needed to enable the positioner control software to communicate seamlessly with the motor controller's accessory module, ensuring real-time command and data transfer for precise antenna alignment with moving satellites. Lastly, the SDR software interface must facilitate signal reception, interpretation, and frequency adjustments while offering a user-friendly display of received signals and data on a computer screen. These interface requirements are fundamental to the system's efficient and effective operation.

<Describe the requirements associated with any communications functions required by this product, including e-mail, web browser, network server communications protocols, electronic forms, and so on. Define any pertinent message formatting. Identify any communication standards that will be used, such as FTP or HTTP. Specify any communication security or encryption issues, data transfer rates, and synchronization mechanisms.>

4. System Features

Without more information from the product owner, we are unsure of how to categorize our system's features

<This template illustrates organizing the functional requirements for the product by system features, the major services provided by the product. You may prefer to organize this section by use case, mode of operation, user class, object class, functional hierarchy, or combinations of these, whatever makes the most logical sense for your product.>

4.1 System Feature 1

<Don't really say "System Feature 1." State the feature name in just a few words.>

4.1.1 Description and Priority

<Provide a short description of the feature and indicate whether it is of High, Medium, or Low priority. You could also include specific priority component ratings, such as benefit, penalty, cost, and risk (each rated on a relative scale from a low of 1 to a high of 9).>

4.1.2 Stimulus/Response Sequences

<List the sequences of user actions and system responses that stimulate the behavior defined for this feature. These will correspond to the dialog elements associated with use cases.>

4.1.3 Functional Requirements

<Itemize the detailed functional requirements associated with this feature. These are the software capabilities that must be present in order for the user to carry out the services provided by the feature, or to execute the use case. Include how the product should respond to anticipated error conditions or invalid inputs. Requirements should be concise, complete, unambiguous, verifiable, and necessary. Use "TBD" as a placeholder to indicate when necessary information is not yet available.>

<Each requirement should be uniquely identified with a sequence number or a meaningful tag of some kind.>

REQ-1:
REQ-2:

4.2 System Feature 2 (and so on)

5. Other Nonfunctional Requirements

5.1 Performance Requirements

5.1.1 The system shall track the given satellite in real time
5.1.2 The system shall adjust to moving position of the given satellite
5.1.3 The system shall TX/RX at the given frequencies at all times operation
5.1.4 The system shall perform adjusting movements without interfering with communications
5.1.5 The system shall only user the necessary power required

These requirements exist so that the system will perform satellite tracking efficiently and effectively. The goal is for the system to adjust to a satellite's ever changing position and perform the necessary actions to consistently track them. The system also needs to accurately transmit and receive data because otherwise the operations would not be possible.

5.2 Safety Requirements

5.2.1 The system shall cease operation in any event that'd cause damage to components
5.2.2 The system shall cease operation if it interferes with local communications
5.2.3 The system shall cease movement operations if the movement can not be performed(obstacles)
5.2.4 The system shall perform consistent movements over time
5.2.5 The system shall not overheat during its operation
5.2.6 The system shall not draw power while not in operation

5.3 Security Requirements

5.3.1 The system shall not interfere will local communications
5.3.2 The system shall not integer will local laws regarding communication equipment
5.3.3 The system shall be accessible by only the user
5.3.4 The system shall not interfere in local first responder operations
5.3.5 The system shall not interfere with the satellites or other satellite operations
5.3.6 The system shall have a passcode for the user interface

5.4 Software Quality Attributes

5.4.1 The software will perform adjusting movements to near 100% accuracy
5.4.2 The software will perform active tracking via the given frequencies and locations
5.4.3 The software will have reboot capabilities in the events of power cycling and or errors
5.4.4 The software will have adjustable operations in the event of loss communications

5.5 Business Rules

5.5.1 There will be one operator of the vehicle controlling the system via software and or a mechanical adjustor
5.5.2 System use approval shall be given before the commencement of operations
5.5.3 System use should be known to all parties involved

6. Other Requirements

We do not yet have other requirements for this product
<Define any other requirements not covered elsewhere in the SRS. This might include database requirements, internationalization requirements, legal requirements, reuse objectives for the project, and so on. Add any new sections that are pertinent to the project.>

Appendix A: Glossary

To be completed as a more definitive draft is made
<Define all the terms necessary to properly interpret the SRS, including acronyms and abbreviations. You may wish to build a separate glossary that spans multiple projects or the entire organization, and just include terms specific to a single project in each SRS.>

Appendix B: Analysis Models

To be added once design analysis has been completed
<Optionally, include any pertinent analysis models, such as data flow diagrams, class diagrams, state-transition diagrams, or entity-relationship diagrams.>

Appendix C: To Be Determined List

<Collect a numbered list of the TBD (to be determined) references that remain in the SRS so they can be tracked to closure.>